

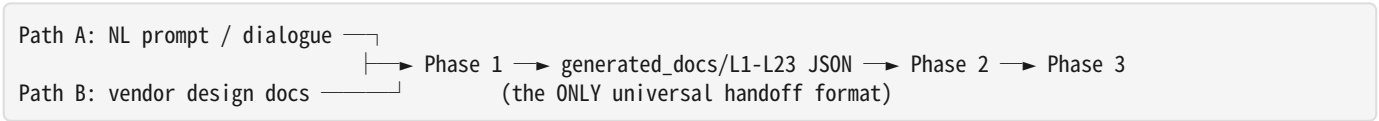
Vibe-IC Canonical Flow v2.2.0

Plugin version: 0.2.2 • **Supersedes:** CANONICAL_FLOW_v2.0.0.md, PHASE_VS_STAGE_VS_INDUSTRY_TAXONOMY.md, docs/tutorials/33_step_flow_overview.md, docs/design/STANDARD_FLOW.md (all stop at the 33-step / 54-entity model and predate the 81-protocol Phase-1 dispatch, the Phase-2 scaffold bridge, and the doc→GDS LVS sign-off chain).

Source of truth = the runners, not this prose. Every step below is cited to the actual step marker in its *_one_shot_runner.py. When a runner changes, regenerate this doc from the markers (grep -noE '\[[0-9]+\][a-z0-9]*/[0-9]+\]' per runner) — do not hand-drift it.

0. Entry paths + orchestrator

programs/vibe_ic_one_shot_runner.py is the top orchestrator. Two entry paths, one handoff:



Orchestrator sequence (vibe_ic_one_shot_runner.py): **Phase 1 → Phase 2 (=2a+2b) → Analog (after Phase 2, so L5_ADI_SPEC is populated; non-blocking) → Phase 3.** FAIL gating: Phase 1 FAIL halts before 2; Phase 2 FAIL halts before 3; Phase 3 FAIL → verdict FAIL but report emitted.

Phase	Transform	Runner	Gate
P0 pre-flight	env / PDK / tool availability	mcp_server_health_check, eda_doctor	tools reachable
Phase 1	any input → L1-L23 JSON (+ human MD)	phase1_doc_one_shot_runner.py	24 L-docs + completeness/parity
Phase 2	L1-L23 → verified RTL → (gate netlist / FPGA SOF)	phase2_one_shot_runner.py	lint/synth/conformance/TB + final_audit
Analog A1-A8	L1/L5 → sized block → hardmacro (parallel to P2)	analog_one_shot_runner.py	per-block DRC/LVS + corners
Mixed-signal	digital + analog co-sim	skill mixed-signal-cosim (no dedicated runner)	M1-M4 co-sim
Phase 3	netlist → synth → PnR → GDS → DRC → LVS	phase3_one_shot_runner.py	DRC 0 + LVS + STA + tapeout checklist

1. Phase 1 — 15 steps (phase1_doc_one_shot_runner.py, the [N/15] markers)

Marker	Step	Output
[1/15]	Extract text from input/docs/ (PDF/DOC/XLSX/...)	input_doc/ (capped at 2 MB/doc for the O(n²) scans, v0.1.91)
[2/15]	L1_DATASHEET	generated_docs/L1_DATASHEET.json
[3/15]	L2_FRS	L2
[4/15]	L3_CMD_PROTOCOL	L3
[5/15]	L4_REGMAP	L4
[6/15]	L5_ADI_SPEC	L5
[7/15]	L6_CONTROL_LOGIC	L6
[8/15]	L7_TEST_DEBUG	L7

Marker	Step	Output
[9/15]	L8_RTL_CONSTANTS	L8_RTL_CONSTANTS
[10/15]	L9_INTEGRATION_SPEC	L9
[11/15]	L10_TEST_CASES	L10
[12/15]	L11_OTP_CONTENT	L11
[13/15]	L12_BEHAVIORAL_SEQUENCES	L12
[14/15]	L13_LAB_CALIBRATION	L13
[14b/15]	L8_TIMING_WAVEFORM (+ 14b2 width, 14b3 encoding, 14b7 universal constants, 14b4 L6 FSM, 14b5 L12 seq)	L8_TIMING_WAVEFORM + overlays
[14c/15]	L14-L18 protocol spec extract (+ 14c0 L9, 14c1 L1 meta, 14c1b L17 handshake, 14c2 L3 mirror, 14c3 L17/L18/L8_TIMING/L9 batch synth, 14c4 universal doc facts, 14c5 residual cleanup)	L14-L18
[14d/15]	L19-L23 skeleton emit	L19-L23
[14e/15]	serial_peripheral_protocol_class synth (R53/R54/R55) — the protocol detector→synth dispatch	per-protocol L-doc overlays
[14e2/15]	bus_interconnect_protocol Tier-2 synth (TileLink/Wishbone/Avalon/OCP/AXI-Stream)	bus-protocol overlays
[14e3/15]	Universal packet/PDU L10↔L3 opcode-consistency sweep	L10 cleaned
[15/15]	Coverage / parity report	reports/phase1/

The [14e/15] block is the 81-protocol-class engine (this session, v0.1.84→v0.2.2). Each protocol ships a content-only is_<proto>(blob) detector + <proto>_protocol_synth.py overlay, dispatched by ic_class. Families: serial (SPI/I2C/I3C/UART/1-Wire/JTAG/SWD/QSPI/SMBus/MIPI-SPMI-RFFE), automotive (CAN/CAN-FD/LIN/FlexRay/SENT/PSI5/Modbus/RS-485/CANopen), memory (DDR3/4/5/LPDDR5/HBM3/GDDR6/ONFI/ eMMC/SD-MMC/UFS/NVMe/HyperBus), display (HDMI/MIPI/DSI/CSI-2/DisplayPort/eDP), PCIe family (PCIe/Gen5/CXL/NVLink/UCIe), USB (2.0/USB4), networking (Ethernet/800G/HDLC/SpaceWire/AFDX/Auto-Eth/ InfiniBand/FibreChannel/PROFIBUS/PROFINET/IO-Link/EtherCAT), wireless (BLE/NFC/Zigbee/LoRa), audio (I2S/SoundWire/S-PDIF/A2B), data-converter (JESD204), debug (CoreSight), timing (PTP), aerospace (MIL-STD-1553/ARINC429), bus (AHB-APB/AXI-ACE-CHI/TileLink/Wishbone/Avalon/OCP/AXI-Stream), security (TPM).

Guard: every module-level detector is auto-covered by tests/test_protocol_detector_no_misfire.py (no foreign-benchmark fire; derived-sibling allowlist in protocol_detector_lib.DERIVED_SIBLING_CROSS_FIRES).

2. Phase 2 — RTL authoring + verification (phase2_one_shot_runner.py , step_*)

Order	Step (def_step_*)	Notes
1	step_phase1	re-run/ingest Phase 1 if needed
2	step_rig_topology_skeleton	scaffold topology
3	step_rtl_gen	WAIVES for ic_class with rtl_gen=null → AI fills the spec-to-rtl role inside the pipeline (digital_arithmetic_primitive / digital_cmd_driven / serial_peripheral_protocol / bus_interconnect_protocol / processor_cpu / unknown). phase2_scaffold_gen.py emits top/regs/fsm/tb/soc_wrap/cocotb scaffold deterministically.
4	step_full_stack_tb_gen	self-checking TB generation

Order	Step (def_step_*)	Notes
5	step_reference_tb	reference-TB conformance (eco_loop up to 3 retries)
6	step_yosys_synth	gate-level synth
7	step_qsf_gen / step_sdc_gen	FPGA project + constraints
8	step_otp_image_check	OTP image (if applicable)
9	step_fpga_compile	Quartus/yosys FPGA build → .sof
10	step_fpga_burn	program board (hardware path)
11	step_usb_hid_tester_verify	host-side protocol-tester acceptance
12	step_emit_phase2_manifests	phase2 manifests
13	step_final_audit	aggregate audit gate

Surrounding deterministic gates (fire around the AI authoring step): rtl_hygiene_lint --fix, spec_conformance_check, chip_top_gate_wrapper_gen, MCP_eda_lint / eda_synth / eda_cocotb.

3. Analog A1-A8 (analog_one_shot_runner.py , parallel to Phase 2)

Step	Name	Output
A1	spec_extract	analog/<block>/A1_spec.json
A2	topology_select	A2_topology.json
A3	netlist_gen	<block>.sp
A4	corner_sweep	A4_corners.json
A5	layout (Magic)	A5_layout.json (needs DRC-clean + LVS-match flags)
A6	post_layout_resim	A6_postsim.json
A7	hardmacro_gen	{.lef, .lib, .gds, .v} → feeds Phase 3
A8	hw_verify (HIL)	A8_hw_verify.json

(Older docs listed "A1-A9"; the runner ships **A1-A8**. Mixed-signal **M1-M4** is skill-level mixed-signal-cosim, no dedicated *_one_shot_runner.py.)

4. Phase 3 — physical design + sign-off (phase3_one_shot_runner.py , step_*)

Step	Name	Tool (open-source substitute)
1	step_synth	yosys (sky130/gf180) — + tie-cell pass (see § 5)
2	step_pnr	OpenROAD (floorplan → PDN → place → CTS → route)
3	step_gds	KLayout streamout (def2gds ; OpenROAD no longer streams GDS)
4	step_drc	KLayout sky130 deck
5	step_lvs	netgen / yosys_equiv — the sign-off chain in § 5
6	step_canonicalize_artefacts	normalize outputs

Tapeout gate: tapeout-checklist (DRC/LVS/STA/IR/EM/antenna/ERC/LEC/DFT) + flow_compliance_check.

5. The LVS sign-off chain — NEW in this era (v0.1.96 → v0.2.2)

This is the part **entirely missing from the old docs**. `step_lvs` is no longer a single `yosys_equiv` call; it is a layered chain whose every layer was hardened by the 7 doc→GDS pilots:

1. **Structural LEC (default)** — `eda_lvs mode=yosys_equiv (equiv_simple+equiv_induct)`. Residual "unproven" cells = yosys SAT-model gap on PDK primitives (Category-D tool limit, NOT a mismatch). No yosys flag closes it for all cells.
2. **Device-level coverage** — to cover the SAT gap: `eda_extraction (magic ext2spice) + eda_lvs mode=netgen + lvs_netgen_setup_emit.py` (power-net globalization). Compares transistors → no SAT-model concept. Reaches device-class-exact (HDLC 20937=20937; sha256 12148=12148).
3. **Powered-netlist closure** — OpenROAD `write_verilog -include_pwr_gnd` (after `global_connect`) gives the schematic side real VPWR/VGND/VPB/VNB → eliminates the tie-cell disconnected-node residual.
4. **Top-level port labels** — Route A (canonical): `magic_port_extract_emit.py` (export PDK + port makeall). Route B (fallback): `lvs_def_port_seed.py` (parse DEF PINS). (Route A is required; B is an audit hint — empirically confirmed.)
5. **Sign-off guard (MANDATORY)** — `lvs_signoff_guard.py`: RAISES on a claimed match against a PORTLESS extracted top `.subckt` (the vacuous / silent-false-positive condition). Run before trusting ANY LVS match.

Synth tie-cell pre-step (§ 4 step 1): `setundef -zero; hilomap -hicell conb_1 HI -locell conb_1 L0; splitnets; clean` (NO `opt_clean` — it deletes tie cells). Without it, constant nets fail TritonRoute DRT-0305. The bare MCP `eda_synth` path lacks this (backlog ORGANIC-20260531-mcp-eda-synth-missing-hilomap-tiecells); `phase3_one_shot_runner` does it automatically. Forward-validated SENT→QSPI→HDLC→SpaceWire.

5b. Phase-3 sign-off checks — gap status

`step_drc` / `step_lvs` are the headline gates, but a complete sign-off runs more checks. The rows below are the **sign-off checks and their status** (not every physical-design step — placement / CTS / routing / output / ECO are covered in § 4 and the § 6 pilots and pass). **None of the gaps below is a circuit-design error** — they are script-ordering / cascading / environment / report-schema issues. Severities + root causes (audited):

Step-number caveat (this is why "step 23" looked missing): two numberings exist. The IDs below follow the **sign-off-audit** scheme (SPEF 22 / STA 23 / IR 24 / EM 25 / Antenna 26 / SI 27 / DRC-LVS-ERC 30 / fill 33). The repo's `33_step_flow_overview.md` numbers the SAME checks differently (SPEF 20 / STA 21 / IR 22 / EM 23 / Antenna 24 / SI 25 / PV 27). Reconcile to one scheme when `flow_doc_emit.py` lands (§ 8). **Step 23 = Post-route STA**, listed below as PASS — it was absent earlier only because the gap-focused draft omitted passing checks.

Step	Check	What it is (one line)	Open-source status	Severity
22	SPEF (OpenRCX)	per-wire R/C "parasitic" extract — feeds STA/IR/EM/SI	<code>extract.tcl</code> must call <code>global_route + set_wire_rc</code> first, else no SPEF	◆ medium (script order)
23	Post-route STA (MMMC)	multi-corner static timing sign-off	runs once SPEF exists; pilots report setup slack +X ns MET (3-corner)	● none (passes)
24	IR drop (PSM)	power-grid voltage-droop under switching current	cascading-missing: unlocks once SPEF (22) exists	◆ medium (cascade)
25	EM	electromigration — current density erodes metal over time	cascading-missing: needs SPEF (22)	◆ medium (cascade)
27	SI	crosstalk — a net's transition couples to neighbours	cascading-missing: needs SPEF (22)	◆ medium (cascade)
26	Antenna			

Step	Check	What it is (one line)	Open-source status	Severity
		long routed metal accumulates plasma charge → gate punch-through	OpenROAD router already runs it; report not on the audit path	● low (done, report-path)
30	DRC / LVS / ERC	foundry rule check / layout-vs-schematic / electrical-rule	sky130 PDK ships only Calibre decks; open-source needs KLayout/ Magic decks wired (this era added the device-level netgen LVS chain in § 5)	● high (env / deck)
33	Metal fill	dummy metal for CMP density uniformity	runner lacks the fill stage → no <code>filled.def</code>	◆ medium (missing stage)
18	Spare cells	tied-off ECO spares for metal-only late fixes	30 spares placed correctly; <code>spare_cells.json</code> missing a <code>rows[]</code> field → audit can't read	● low (report schema)
5	Formal	SAT/model-checking proof (vs sampled simulation)	<code>altsyncram</code> primitive has no formal model → INFORMATIONAL waiver (function covered by post-layout sim, step 28)	● none (waived)

Doctrine: treat each like the LVS chain (§ 5) — separate the design-correctness signal (here: clean) from the tool/script/report signal (here: the actionable gaps). The actionable fixes (SPEF `extract.tcl` ordering; the IR/EM/SI cascade it unblocks; open-source DRC/LVS decks; the metal-fill stage; the `spare_cells.json` schema field) are tracked in `ORGANIC-20260531-phase3-signoff-chain-open-source-gaps`.

6. doc→GDS pilot evidence (7 pilots, real sky130A GDS)

Pilot	Archetype	ic_class	LVS stop point
i2s	streaming-rx	digital_cmd_driven	device-level exact 4499=4499 (3 SAT-unproven → 0; 0 tie cells, port-label floor)
ahb_apb	bus-bridge	bus_interconnect	—
ufs	storage-framer	serial_peripheral	—
sent	sensor-decoder	digital_arithmetic_primitive	structural all-proven 1388/1388
qspi	command-controller	serial_peripheral	structural all-proven 1434/1434
hdlc	packet-framer	digital_cmd_driven	device-level exact 20937=20937 (SAT gap → 0)
spacewire	link credit-flow-control	digital_arithmetic_primitive	device-level exact 6676=6676 / powered 6164=6164 (99 SAT-unproven → 0; port-label floor)

All three device-level-closed pilots (i2s + hdlc + spacewire) now share one stop point — the § 5 LVS chain end-to-end: structural-LEC SAT residual → device-level netgen (covers it to device-class-exact, every SAT-unproven cell → 0) → powered-netlist (eliminates the tie-cell power-pin node; N/A for i2s, which has 0 tie cells) → residual = the Category-D port-label floor (`port makeall` / sign-off LVS), with `lvs_signoff_guard` correctly refusing a vacuous portless match. No pilot now stops at the structural-SAT gap.

7. Project folder layout (unchanged from v2.0.0)

```
<project>/
input/{docs,phase1_prompt.md}      input_doc/      (Phase 1 in/extracted)
phase1/{generated_docs,human_docs,claude_extracted}/L*.json|md
phase2/stage1/{rtl,scaffold,fpga}/  phase2/stage2/synth/
analog/<block>/{A1_spec.json,...,<block>.{sp,lef,lib,gds,v}}
phase3/stage{1..4}/{synth,pnr,gds,drc,lvs}/  phase3/stage5_manufacturing/
reports/{phase1,phase2,phase3,orchestrator}/
```

8. Keeping this current

This doc is **derived**, not authoritative. The live runner-marker step lists are auto-generated by `programs/flow_doc_emit.py` (shipped v0.2.3) into `FLOW_STEPS_GENERATED.md` — the same pattern as `INDEX.md`. Regenerate after any runner change with `python3 flow_doc_emit.py`; a freshness gate (`flow_doc_emit.py --check`, pinned by `tests/test_flow_doc_emit.py`) fails CI if the committed file drifts. This narrative doc + the curated 33-step / LVS-chain / sign-off tables stay hand-authored and link to the generated file.